Ratios, rates, proportional relationships, and units

Problem-Solving and Data Analysis · Math · 93 questions

1 9594986d-588a-4127-87f9-41a6f360b9e1 · Easy

What length, in centimeters, is equivalent to a length of 51 meters?
1 meter = 100 centimeters

A. 0.051

B. 0.51

C. 5,100

D. 51,000

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2 ae2818a7-d027-48ec-a6f6-d9167cb25167 · Easy

A special camera is used for underwater ocean research. When the camera is at a depth of 58 fathoms, what is the camera's depth in feet? 1 fathom = 6 feet

Student-produced response: _____

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3 10654-DC · Easy

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

A. $551.25

B. $562.50

C. $5,625.00

D. $98,000.00

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4 d6f9ca5e-947e-4a5c-a0d0-de08d17d11f5 · Easy

A printer produces posters at a constant rate of 42 posters per minute. At what rate, in posters per hour, does the printer produce the posters?

Student-produced response: _____

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5 029578-DC · Easy

x	y
1	4
3	12
5	20
40	k

In the table above, the ratio of y to x for each ordered pair is constant. What is the value of k ?

A. 28

B. 36

C. 80

D. 160

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6 029322-DC · Easy

For a person m miles from a flash of lightning, the length of the time interval from the moment the person sees the lightning to the moment the person hears the thunder is k seconds. The ratio of m to k can be estimated to be 1 to 5. According to this estimate, the person is how many miles from a flash of lightning if the time interval is 25 seconds?

A. 10

B. 9

C. 6

D. 5

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7 c11c7138-6922-4b01-8ac3-981d24884e18 · Easy

How many meters are equivalent to 2,300 centimeters? 100 centimeters = 1 meter

A. 0.043

B. 23

C. 2,400

D. 230,000

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8 04108-DC · Easy

To make a bakery's signature chocolate muffins, a baker needs 2.5 ounces of chocolate for each muffin. How many pounds of chocolate are needed to make 48 signature chocolate muffins? (1 pound = 16 ounces)

A. 7.5

B. 10

C. 50.5

D. 120

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9 a9c40475-9ebc-41fc-bfa0-50236b060e50 · Easy

An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?

A. 9

B. 96

C. 120

D. 972

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10 bc2db446-6593-43b2-b2b0-2932e35e1153 · Easy

A special camera is used for underwater ocean research. The camera is at a depth of 39 fathoms. What is the camera's depth in feet? $1 \quad \text{fathom} = 6 \quad \text{feet}$

A. 234

B. 117

C. 45

D. 7

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11 05121-DC · Easy

The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. The weight of an object on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, approximately how many more pounds does it weigh on Jupiter than it weighs on Venus?

A. 90

B. 111

C. 140

D. 230

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12 47ff0dd5-480f-4331-b9a9-ee44e9b40d9e · Easy

A fish swam a distance of 5,104 yards. How far did the fish swim, in miles?

1 mile = 1,760 yards

A. 0.3

B. 2.9

C. 3,344

D. 6,864

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13 022849-DC · Easy

A group of monarch butterflies migrated from Chicago, Illinois, to Michoacán, Mexico, flying a total of 2,100 miles. It took a single butterfly in the group 120 days to travel this route one way. On average, how many miles did the butterfly travel per day?

A. 0.057

B. 0.729

C. 17.5

D. 24

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14 d88daa3d-713d-4905-8fc3-540215e519e9 · Easy

In a box of pens, the ratio of black pens to red pens is 8 to 1. There are 40 black pens in the box. How many red pens are in the box?

A. 5

B. 8

C. 40

D. 320

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15 a1a5d2fd-4a30-4105-b4de-ee544853263c · Easy

Tilly earns p dollars for every w hours of work. Which expression represents the amount of money, in dollars, Tilly earns for $39w$ hours of work?

A. $39p$

B. $\frac{p}{39}$

C. $p + 39$

D. $p - 39$

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16 973f54f3-20e6-4be8-a0e8-f04182a1dc93 · Easy

How many yards are equivalent to 77 rods? $5.5 \quad \text{yards} = 1 \quad \text{rod}$

Student-produced response: _____

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17 9ff69db4-88d5-4c3b-ad95-8bd4d0c3f462 · Easy

The ratio x to y is equivalent to the ratio 12 to t. When $x = 156$, what is the value of y in terms of t?

A. $13t$

B. $12t$

C. $144t$

D. $168t$

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18 031618-DC · Easy

If $t = 4u$, which of the following is equivalent to $2t$?

A. $8u$

B. $2u$

C. u

D. $\frac{1}{2}u$

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19 b5f08f84-0852-405a-b44a-c1ad327e11dd · Easy

A product costs 11.00 dollars per pound. What is the cost, in dollars, for 6 pounds of the product?

Student-produced response: _____

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20 f715f428-619d-426d-993d-afd4cdeeb5d5 · Easy

How many teaspoons are equivalent to 44 tablespoons? 3 teaspoons = 1 tablespoon

A. 47

B. 88

C. 132

D. 176

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21 029014-DC · Easy

The ratio of t to u is 1 to 2, and $t = 10$. What is the value of u ?

A. 2

B. 5

C. 10

D. 20

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22 70312974-5b98-4396-b51e-4f590919f407 · Easy

A certain bird species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this bird species fly in 4 seconds?

A. 64

B. 20

C. 16

D. 12

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23 69320716-58db-43ad-b02d-1664b3dfb359 · Easy

The population density of Worthington is 290 people per square mile. Worthington has a population of $92{,}800$ people. What is the area, in square miles, of Worthington?

A. $102{,}400$

B. $93{,}090$

C. 320

D. 32

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24 5979deaa-edcc-42e7-8d12-24a01cec2c7a · Easy

A customer spent $\$\,27$ to purchase oranges at $\$\,3$ per pound. How many pounds of oranges did the customer purchase?

Student-produced response: _____

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25 9d4083e9-df43-4c60-9f48-cb64e2d3dc9d · Easy

A participant in a bicycle race completes the race with an average speed of $24{,}816$ yards per hour. What is this average speed, in miles per hour? $1 \text{ mile} = 1{,}760 \text{ yards}$

Student-produced response: _____

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26 f9b50fe8-25d0-4faa-ad54-7ba7d01af7db · Easy

A cable provider wanted to know how many of its $30{,}000$ customers would be interested in a new service plan. The provider selected 300 customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, 8 said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

A. 8

B. 80

C. 800

D. $8{,}000$

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27 334db357-0b99-4f13-a766-9dde9a462af3 · Easy

An object travels at a constant speed of 6 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 24 centimeters?

Student-produced response: _____

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28 025315-DC · Easy

Makayla is planning an event in a 5,400-square-foot room. If there should be at least 8 square feet per person, what is the maximum number of people that could attend this event?

A. 588

B. 675

C. 15,274

D. 43,200

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29 ee5f324c-581b-4818-b0a7-48d75fb59f22 · Easy

How many yards are equivalent to 612 inches? 1 yard = 36 inches

A. 0.059

B. 17

C. 576

D. 22,032

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30 872cefa5-14e6-4009-b839-96c77f5f3579 · Easy

How many cups are equivalent to 83 quarts? 4 cups = 1 quart

A. 21

B. 79

C. 87

D. 332

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31 c40a706f-ca84-432c-bdb2-58a4920cbaab · Easy

A kangaroo has a mass of 28 kilograms. What is the kangaroo's mass, in grams?
(1 kilogram = 1,000 grams)

A. 28,000

B. 1,028

C. 972

D. 784

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32 a824dc18-95e2-4efa-b635-884db6f2eeef · Easy

A wind turbine completes 900 revolutions in 50 minutes. At this rate, how many revolutions per minute does this turbine complete?

A. 18

B. 850

C. 950

D. 1,400

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33 d6f8cc02-96fe-42b9-bfb6-5bd07411ccfc · Easy

An object's speed is 64 yards per second. What is the object's speed, in feet per second? 1 yard = 3 feet

A. 61

B. 67

C. 94

D. 192

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34 80f12ef5-a97c-4f43-a382-5d490d55eb90 · Easy

A giant armadillo has a mass of 39 kilograms. What is the giant armadillo's mass in grams? (1 kilogram = 1,000 grams)

Student-produced response: _____

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35 313142de-ba5a-45df-9ffe-0993fba11dce · Easy

How many feet are equivalent to 34 yards? 1 yard = 3 feet

Student-produced response: _____

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36 4c11262b-41b4-4c46-84a0-c2f957e8f9e5 · Easy

How many yards are equivalent to 1,116 inches? 1 yard = 36 inches

Student-produced response: _____

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37 688b466f-c743-40b5-8bd7-7b8695af38e7 · Easy

The ratio of the length of line segment *XY* to the length of line segment *ZV* is 6 to 1. If the length of line segment *XY* is 102 inches, what is the length, in inches, of line segment *ZV*?

A. 17

B. 96

C. 102

D. 612

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38 444c8744-ce2f-4537-88b4-072fc2d44492 · Easy

A cluster fly is traveling at a velocity of 76 centimeters per second. What is this velocity, in millimeters per second? 1 centimeter = 10 millimeters

A. 7,600

B. 836

C. 760

D. 86

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39 1aadad19-c839-4ddb-aa09-87de2e47212a · Easy

The number of raccoons in a 131-square-mile area is estimated to be $2{,}358$. What is the estimated population density, in raccoons per square mile, of this area?

A. 18

B. 131

C. 149

D. $2{,}376$

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40 028058-DC · Easy

Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?

A. 1 to 4

B. 4 to 1

C. 1 to 7

D. 7 to 1

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41 c99660ad-c63a-459b-a649-55773967d912 · Easy

A cherry pitting machine pits 12 pounds of cherries in 3 minutes. At this rate, how many minutes does it take the machine to pit 96 pounds of cherries?

A. 8

B. 15

C. 24

D. 36

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42 eae9c7b8-873d-4ff8-92bd-dffd8ccec2b8 · Easy

A mechanical device in a workshop produces items at a constant rate of 60 items per hour. At this rate, how many items will the mechanical device produce in 3 hours?

Student-produced response: _____

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43 7904b1f4-ccaf-441c-ae58-d8dc651e91cc · Medium

There is a proportional relationship between p and t. If $p = 5{,}130$ when $t = 6{,}840$, then what is the value of p when $t = 5{,}472$?

A. $7{,}296$

B. $7{,}182$

C. $4{,}104$

D. $3{,}762$

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44 3d00b62c-b08f-45da-ac4b-7750ae794184 · Medium

An object has a mass of 168 grams and a volume of 24 cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

A. 7

B. 144

C. 192

D. $4{,}032$

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45 72a6fbc6-5106-42d1-b891-58079546b368 · Medium

The total mass, in kilograms, of r identical objects is t. Which expression represents the total mass, in kilograms, of $146r$ of these objects?

A. $146 - t$

B. $146 + t$

C. $\frac{t}{146}$

D. $146t$

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46 14884-DC · Medium

Food	Protein	Cost
1 large egg	6 grams	$0.36
1 cup of milk	8 grams	$0.24

The table above shows the amount of protein in two foods and the cost of each food. Based on the table, what is the ratio of the cost per gram of protein in a large egg to the cost per gram of protein in a cup of milk?

A. 1 : 2

B. 2 : 3

C. 3 : 4

D. 2 : 1

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47 070608-DC · Medium

Pure beeswax has a density of 0.555 ounce per cubic inch. An online company sells pure beeswax at a price of $8.00 per ounce. What is the selling price, in dollars per cubic inch, for pure beeswax purchased from this company?

Student-produced response: _____

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48 c53c49b5-ed37-4cd3-b08c-7994919f7e09 · Medium

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y ?

Student-produced response: _____

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49 252ab91d-897c-4c0d-8d50-e60456359300 · Medium

For a certain rectangular region, the ratio of its length to its width is 35 to 10. If the width of the rectangular region increases by 7 units, how must the length change to maintain this ratio?

A. It must decrease by 24.5 units.

B. It must increase by 24.5 units.

C. It must decrease by 7 units.

D. It must increase by 7 units.

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50 27dab70d-b227-496d-9793-7bbb537c6638 · Medium

For the values j and k, the ratio of j to k is 11 to 12. If j is multiplied by 17, what is k multiplied by in order to maintain the same ratio?

Student-produced response: _____

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51 e2e5fe90-eac5-48a3-a466-f4a6ead5776e · Medium

A distance of 112 furlongs is equivalent to how many feet?

1 furlong $= 220$ yards and 1 yard $= 3$ feet

Student-produced response: _____

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52 98ff2d18-fd0b-4125-ae68-dffac04ee1ef · Medium

The ratio *a* to *b* is equivalent to the ratio 63 to 17. What is the value of *a* when $b = 153$?

Student-produced response: _____

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53 d2f59729-f3aa-4681-adb0-ddf2309976b8 · Medium

A competition consisted of four different events. One participant completed the first event with an average speed of 20.300 miles per hour. What was this average speed, in yards per hour? $1 \text{ mile} = 1{,}760 \text{ yards}$

Student-produced response: _____

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54 65c97795-cef4-4246-9fe9-82f67f8f7823 · Medium

One side of a flat board has an area of 874 square inches. If a pressure of 19 pounds per square inch of area is exerted on this side of the board, what is the total force, in pounds, exerted on this side of the board?

Student-produced response: _____

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55 028848-DC · Medium

Rectangle A has length 15 and width w. Rectangle B has length 20 and the same length-to-width ratio as rectangle A. What is the width of rectangle B in terms of w ?

A. $\frac{4}{3}w$

B. $w+5$

C. $\frac{3}{4}w$

D. $w-5$

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56 021174-DC · Medium

Tanya earns $13.50 per hour at her part-time job. When she works z hours, she earns $13.50z$ dollars. Which of the following expressions gives the amount, in dollars, Tanya will earn if she works $3z$ hours?

A. $3(13.50z)$

B. $3+13.50z$

C. $3z+13.50z$

D. $13.50(z+3)$

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57 bc6c4c12-4c9a-4382-963e-957612ede87c · Medium

A distance of 354 furlongs is equivalent to how many feet?

1 furlong = 220 yards and 1 yard = 3 feet

A. 306

B. 402

C. 25,960

D. 233,640

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58 e098ab1f-ca25-4df4-862e-fe8448392a67 · Medium

An insect moves at a speed of $\frac{3}{20}$ feet per second. What is this speed, in yards per second?

3 feet = 1 yard

A. $\frac{1}{20}$

B. $\frac{9}{20}$

C. 6

D. 20

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59 e7ba7b4b-ddc3-498e-9508-402688a85aef · Medium

The ratio 140 to m is equivalent to the ratio 4 to 28. What is the value of m?

Student-produced response: _____

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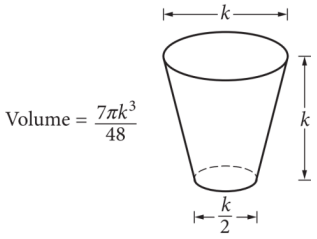
60 01f43331-d478-491e-90be-1bf336840ec3 · Medium

The population density of Cedar County is 230 people per square mile. The county has a population of $85{,}100$ people. What is the area, in square miles, of Cedar County?

Student-produced response: _____

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61 070902-DC · Medium



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. Jenny has a pitcher that contains 1 gallon of water. How many times could Jenny completely fill the glass with 1 gallon of water?
(1 gallon = 128 fluid ounces)

A. 16

B. 8

C. 4

D. 3

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62 d4996faa-e699-48e5-8816-df8bd4c0420c · Medium

How many fluid ounces are equivalent to 76 quarts?

8 fluid ounces = 1 cup and 4 cups = 1 quart

Student-produced response: _____

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63 295c177c-ee26-49e1-8364-8c9b6677b31a · Medium

If $\frac{x}{y} = 4$ and $\frac{24x}{ny} = 4$, what is the value of n?

Student-produced response: _____

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64 029306-DC · Medium

Which of the following speeds is equivalent to 90 kilometers per hour? (1 kilometer = 1,000 meters)

A. 25 meters per second

B. 32 meters per second

C. 250 meters per second

D. 324 meters per second

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65 10619-DC · Medium

The total area of a coastal city is 92.1 square miles, of which 11.3 square miles is water. If the city had a population of 621,000 people in the year 2010, which of the following is closest to the population density, in people per square mile of land area, of the city at that time?

A. 6,740

B. 7,690

C. 55,000

D. 76,000

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66 021593-DC · Medium

The population density of Iceland, in people per square kilometer of land area, increased from 2.5 in 1990 to 3.3 in 2014. During this time period, the land area of Iceland was 100,250 square kilometers. By how many people did Iceland's population increase from 1990 to 2014?

A. 330,825

B. 132,330

C. 125,312

D. 80,200

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67 cefd9f5a-cd0c-4e0f-b298-90f990380e14 · Medium

At a particular track meet, the ratio of coaches to athletes is 1 to 26. If there are x coaches at the track meet, which of the following expressions represents the number of athletes at the track meet?

A. $\frac{x}{26}$

B. $26x$

C. $x + 26$

D. $\frac{26}{x}$

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68 7d9689ed-c13d-4213-a6f7-685ae6b402ee · Medium

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of 19.700 miles per hour. What was the average speed, in yards per hour, of the participant during the cycling leg?

$1 \text{ mile} = 1{,}760 \text{ yards}$

Student-produced response: _____

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69 019681-DC · Medium

The International Space Station orbits Earth at an average speed of 4.76 miles per second. What is the space station's average speed in miles per hour?

A. 285.6

B. 571.2

C. 856.8

D. 17,136.0

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70 b32d7197-2b76-4429-8221-4e5482d511ae · Medium

One of a planet's moons orbits the planet every 252 days. A second moon orbits the planet every 287 days. How many more days does it take the second moon to orbit the planet 29 times than it takes the first moon to orbit the planet 29 times?

Student-produced response: _____

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71 46ab20b9-f632-4d8f-a085-104d8f446b46 · Medium

How many tablespoons are equivalent to 14 teaspoons? 3 teaspoons $= 1$ tablespoon

Student-produced response: _____

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72 e2565394-6b89-4d8e-8b02-06a363d8a20f · Medium

A distance of 61 furlongs is equivalent to how many feet?
1 furlong $= 220$ yards and 1 yard $= 3$ feet

Student-produced response: _____

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73 028911-DC · Medium

$d = 55t$

The equation above can be used to calculate the distance d, in miles, traveled by a car moving at a speed of 55 miles per hour over a period of t hours. For any positive constant k, the distance the car would have traveled after $9k$ hours is how many times the distance the car would have traveled after $3k$ hours?

A. 3

B. 6

C. $3k$

D. $6k$

---

74 070624-DC · Medium

On April 18, 1775, Paul Revere set off on his midnight ride from Charlestown to Lexington. If he had ridden straight to Lexington without stopping, he would have traveled 11 miles in 26 minutes. In such a ride, what would the average speed of his horse have been, to the nearest tenth of a mile per hour?

Student-produced response: _____

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75 16d02cf9-fae4-4096-a277-a897cfed31be · Hard

A rectangle has an area of 3,168 square inches. What is the area, in square feet, of this rectangle? 1 foot = 12 inches

A. 22

B. 56.3

C. 264

D. 675.4

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76 93a3d90a-8e47-405b-bf18-8a2d7da11a63 · Hard

On average, a certain plant grows 87 millimeters every m months. At this rate, which expression represents the number of millimeters, on average, the plant grows every k years?

A. $\frac{29m}{4k}$

B. $\frac{29k}{4m}$

C. $\frac{1{,}044m}{k}$

D. $\frac{1{,}044k}{m}$

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77 3a321846-3dc0-4a6d-82e6-2cfcf81ef623 · Hard

The area of a rectangular region is increasing at a rate of 250 square feet per hour. Which of the following is closest to this rate in square meters per minute? Use 1 meter = 3.28 feet.

A. 0.39

B. 1.27

C. 13.67

D. 23.24

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78 933baf7b-cfc3-42f9-a9f8-f184d6f97c27 · Hard

The density of a certain type of wood is 353 kilograms per cubic meter. A sample of this type of wood is in the shape of a cube and has a mass of 345 kilograms. To the nearest hundredth of a meter, what is the length of one edge of this sample?

A. 0.98

B. 0.99

C. 1.01

D. 1.02

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79 d6f5d386-dd70-4e85-96cc-2c0859c39acc · Hard

The density of a certain solid substance is 813 kilograms per cubic meter. A sample of this substance is in the shape of a cube, where each edge has a length of 0.60 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

A. 176

B. 488

C. $1{,}355$

D. $3{,}764$

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80 025175-DC · Hard

Anita created a batch of green paint by mixing 2 ounces of blue paint with 3 ounces of yellow paint. She must mix a second batch using the same ratio of blue and yellow paint as the first batch. If she uses 5 ounces of blue paint for the second batch, how much yellow paint should Anita use?

A. Exactly 5 ounces

B. 3 ounces more than the amount of yellow paint used in the first batch

C. 1.5 times the amount of yellow paint used in the first batch

D. 1.5 times the amount of blue paint used in the second batch

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81 ea65c30d-d2d3-4c66-b79e-881cd28208cf · Hard

A landscaper uses a hose that puts $88x$ ounces of water in a bucket in $5y$ minutes. Which expression represents the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate?

A. $\frac{9x}{440}$

B. $\frac{440x}{9}$

C. $\frac{5x}{792}$

D. $\frac{792x}{5}$

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82 964d5413-e030-4000-a612-54941886f548 · Hard

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

A. $12y$

B. $3y$

C. $16y$

D. $6y$

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83 c084b0a0-2744-443d-9e6a-f230e22fdc88 · Hard

To study the moisture content in a group of trees, samples from the trunk of each tree were taken from 25 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 2.56 grams, what is the density of this cube, in grams per cubic centimeter?

Student-produced response: _____

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84 cd36e81a-3f06-40da-ab13-b7cd41cd0e9b · Hard

A sample of oak has a density of 807 kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of 0.90 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

A. 588

B. 726

C. 897

D. $1{,}107$

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85 03862-DC · Hard

Jeremy deposited x dollars in his investment account on January 1, 2001. The amount of money in the account doubled each year until Jeremy had 480 dollars in his investment account on January 1, 2005. What is the value of x ?

Student-produced response: _____

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86 86610db9-1732-40a7-a795-6d71991b27f8 · Hard

To study fluctuations in composition, samples of pumice were taken from 29 locations and cut in the shape of a cube. The length of the edge of one of these cubes is 3.000 centimeters. This cube has a density of 0.230 grams per cubic centimeter. What is the mass of this cube, in grams?

Student-produced response: _____

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87 837ca3a4-1a14-44b1-84b5-5f2a706e30f9 · Hard

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of n?

Student-produced response: _____

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88 ec4a70f3-71cf-4dca-9776-c588830c5d34 · Hard

In a scale model of an office building, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x?

A. $\frac{3x}{1712}$

B. $\frac{17x}{123}$

C. $\frac{3x}{17}$

D. $\frac{17x}{3}$

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89 a9ed5978-3ec1-4d21-966b-82a5c7c417f3 · Hard

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field's strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is 3 times the side length, in meters, of the smaller square. An electric field with strength 29.00 volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is $4{,}640 \quad \text{volts} \cdot \text{meters}$, what is the electric flux, in $\text{volts} \cdot \text{meters}$, of the electric field through the larger square?

Student-produced response: _____

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90 8452380b-8279-44a6-98f5-7b0beb6adc6e · Hard

A certain town has an area of 4.36 square miles. What is the area, in square yards, of this town? $1 \quad \text{mile} = 1{,}760 \quad \text{yards}$

A. 404

B. $7{,}674$

C. $710{,}459$

D. $13{,}505{,}536$

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91 74b99987-9634-407d-adbe-a305f02a1c5f · Hard

An area of 46.00 square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k? $1 \quad \text{nautical} \quad \text{mile} = 1.852 \quad \text{kilometers}$

Student-produced response: _____

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92 56513b01-fe94-443b-8898-f5643702eab1 · Hard

A certain park has an area of 11,863,808 square yards. What is the area, in square miles, of this park? 1 mile = 1,760 yards

A. 1.96

B. 3.83

C. 3,444.39

D. 6,740.8

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93 fc00704d-6b53-445f-a379-309a3e911067 · Hard

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth?
Use 1 mile = 1,609 meters.

A. 0.3

B. 16.3

C. 195.8

D. 220.4
